EVALUATION OF

Industrial Alkali Waste

as a Stabilizing Agent for Soil

[Red Mud / Bauxite Residue — Bayer Process Waste]

B.Tech Minor Project · Department of Civil Engineering · Bakhtiyarpur College of Engineering

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01

Introduction & Background

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INTRODUCTION

Understanding the Need for Soil Stabilization

> *"Over 40% of soils in Indo-Gangetic plains are too weak to support infrastructure without treatment."*

WHAT IS SOIL STABILIZATION?

- Process of improving engineering properties of soil for safe construction
- Enhances shear strength, bearing capacity and stiffness
- Reduces plasticity, swelling and compressibility
- Can be mechanical, chemical or biological in nature
- Essential for road subgrades, embankments and foundations

WHY IS IT NEEDED? (PROBLEM)

- Low bearing capacity — cannot support heavy loads
- High plasticity — swells & shrinks with moisture changes
- Excessive settlement — causes structural damage
- High compressibility — poor foundation soil
- Fine-grained soils dominant in Bihar's Ganga plains

ABOUT RED MUD (INDUSTRIAL ALKALI WASTE)

Bauxite Residue from the Bayer Process — Al_2O_3 Extraction

WHAT IS RED MUD?

- Industrial by-product of the Bayer process (alumina refining)
- Generated during extraction of aluminium oxide (Al_2O_3) from bauxite ore
- Highly alkaline residue — pH range 10 to 13
- Worldwide generation: ~150 million tonnes per year
- Majority is landfilled or stored in tailing ponds — serious environmental hazard
- India: major producers include NALCO, HINDALCO, BALCO

CHEMICAL COMPOSITION

Fe_2O_3	30–60%	*Gives red colour*
Al_2O_3	10–20%	*Pozzolanic activity*
SiO_2	5–15%	*Strength gain*
CaO	2–8%	*Cementitious*
TiO_2	3–10%	*Inert filler*
Na_2O	2–10%	*Alkalinity source*

WHY USE AS STABILIZER?

- Fe/Al/Ca oxides → pozzolanic & cementitious reactions
- Improves strength, reduces plasticity
- Converts waste to a valuable resource

OBJECTIVES OF THE STUDY

Research Goals and Test Matrix

MAIN OBJECTIVE:

To evaluate the effect of industrial alkali waste (Red Mud) as a stabilizing agent for fine-grained subgrade soil by conducting Grain Size Analysis, Standard Proctor Compaction Test and Unconfined Compression Strength (UCS) tests at varying Red Mud proportions.

01

Grain Size Analysis

Determine particle size distribution. Classify soil type (gravel/sand/silt/clay) as per IS:2720 Part 4.

02

Standard Proctor Test

Find Optimum Moisture Content (OMC) and Maximum Dry Density (MDD) for natural soil and Red Mud mixes. IS:2720 Part 7.

03

UCS Testing (0–20% Red Mud)

Measure unconfined compressive strength. Compare qu and Cu values across all proportions. IS:2720 Part 10.

04

Environmental & Economic Evaluation

Assess eco-benefits of Red Mud utilization, waste reduction potential and cost-effectiveness vs lime/fly ash.

MATERIALS USED

Soil · Red Mud · Water

NATURAL SOIL

- Source: Bank of Ganga River, Bihar
- Type: Fine-grained silty/clayey soil
- Collected from 0.5–1.0 m depth
- Air-dried and sieved (4.75 mm IS sieve)
- Weak subgrade — low bearing capacity
- Dominant in Indo-Gangetic flood plains

RED MUD (ALKALI WASTE)

- Source: Aluminium plant (Bayer process)
- Highly alkaline (pH: 10–13) — alkali waste
- Rich in Fe_2O_3, Al_2O_3, SiO_2, CaO
- Proportions tested: 0%, 5%, 10%, 15%, 20%
- Mixed by dry weight of soil
- Eco-friendly utilization of industrial waste

WATER

- Distilled / clean tap water
- Added at Optimum Moisture Content (OMC)
- OMC determined by Standard Proctor Test
- OMC for natural soil ≈ 21.66%
- Ensures uniform sample preparation
- Controlled lab-temperature conditions

EQUIPMENT USED

Laboratory Apparatus for All Tests

01

Sieve Set

IS sieves: 4.75, 2.0, 1.0, 0.425, 0.075 mm for grain size analysis

02

Weighing Balance

Electronic balance — accurate mass measurement to 0.01 g precision

03

Proctor Apparatus

Standard Proctor mould Ø100 mm × h127.4 mm + 2.5 kg rammer, 300 mm drop

04

UCS Machine

Strain-controlled loading frame with proving ring for axial compression testing

05

Mixing Trays & Spatulas

Steel trays and tools for uniform soil-Red Mud blending at target moisture

06

Cylindrical Moulds

Ø37.5 mm × L75 mm PVC moulds for UCS specimen preparation (L/D = 2)

EXPERIMENTAL METHODOLOGY

5-Step Laboratory Workflow — IS:2720

STEP 1

Soil Sampling & Preparation

Collect soil from Ganga bank. Air-dry, break clods, pass through 4.75 mm sieve. Store in airtight bags.

STEP 2

Baseline Tests (Natural Soil)

Conduct Proctor & UCS tests on natural soil (0% Red Mud) to establish control/reference values.

STEP 3

Sample Mix Preparation

Blend soil + Red Mud (5%, 10%, 15%, 20% by dry weight). Add water at OMC for each mix proportion.

STEP 4

UCS Testing (Stabilized Mixes)

Form cylindrical specimens Ø37.5 mm × 75 mm at OMC. Load at 1.25 mm/min. Record peak stress.

STEP 5

Data Analysis & Conclusions

Plot qu vs Red Mud %. Compare with control. Find optimal % giving maximum strength improvement.

IS:2720 Part 4 (Grain Size) · IS:2720 Part 7 (Proctor) · IS:2720 Part 10 (UCS) · IS:2720 Part 12 (Atterberg Limits)

GRAIN SIZE ANALYSIS

IS:2720 Part 4 — Particle Size Distribution

TEST PROCEDURE (IS:2720 Part 4)

- Oven-dry soil sample taken for analysis
- pass through 4.75-mm IS sieve
- Sieve set arranged in descending order of sieve size
- Mechanical shaking for 10 minutes (sieve shaker)
- Weight retained on each sieve recorded
- Cumulative % passing computed

Sieve Sizes Used: 4.75 mm · 2.0 mm · 1.0 mm · 0.425 mm · 0.075 mm (75 µm)

Lab Photo: Soil being washed through 0.075 mm) sieve

STANDARD PROCTOR COMPACTION TEST

IS:2720 Part 7 — Natural Soil (0% Red Mud)

MOULD SPECIFICATIONS

Diameter (d):	100 mm
Height (h):	127.4 mm
Volume (V):	1000 cc
Hammer Mass:	2.6 kg
Free Fall Height:	310 mm
Layers × Blows:	3 × 25

OMC

21.66%

Optimum Moisture Content

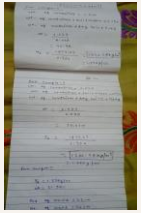
MDD

1.539 g/cc

Maximum Dry Density

Sample	Water Content (%)	Bulk Density γt (kg/m³)	Dry Density γd (g/cc)
1	21.66	1877.27	**1.539 ← OMC & MDD**
2	36.40	1870.00	1.376
3	35.41	1885.00	1.386
4	15.20	1691.00	1.468
5	8.50	1456.00	1.342

** Sample 1 gives OMC = 21.66% and MDD = 1.539 g/cc | Additional tests at higher w% pending to confirm peak*

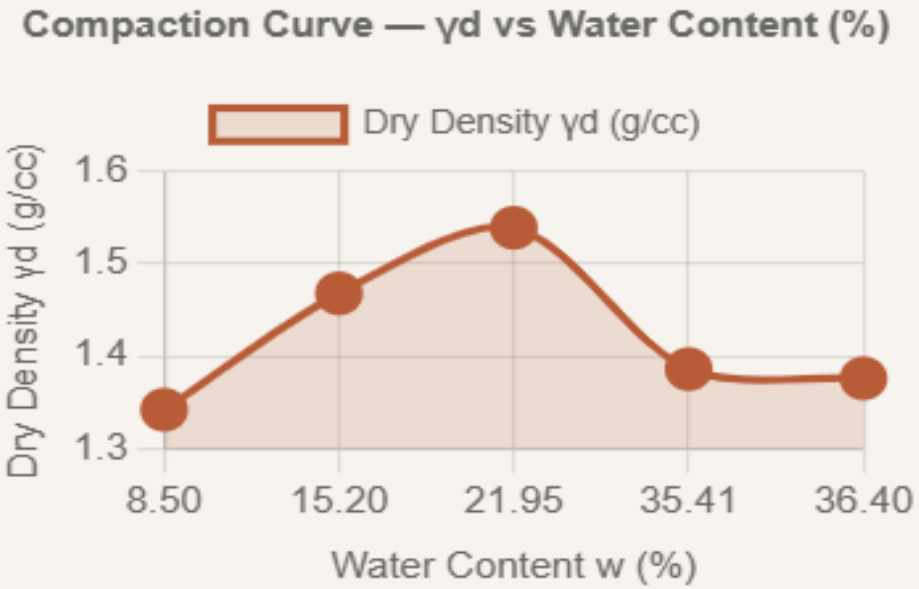


Lab Photos: Standard Proctor compaction test observations

COMPACTION CURVE — NATURAL SOIL

IS:2720 Part 7 · γd vs Water Content — Bell-Shaped Curve

OMC (Peak)	MDD (Peak)	Mould Volume	Standard
21.66%	**1.539 g/cc**	**1000 cc**	**IS:2720-7**
Optimum Moisture Content	*Maximum Dry Density*	*Ø 100mm × h127.4 mm*	*3 layers · 25 blows · 2.5 kg*



Note: Bell-shaped curve rises to peak at OMC = 21.66% then descends — confirming correct Proctor test behaviour

UNCONFINED COMPRESSION STRENGTH (UCS) TEST

IS:2720 Part 10 — Test Setup & Specimen Details

TEST SETUP & SPECIMEN DETAILS	
Specimen Diameter (d):	**37 mm (37.5 mm nominal)**
Specimen Length (L):	**75 mm**
L/D Ratio:	**2.0 (IS standard requirement)**
Initial Cross-Sectional Area (A_o):	**1075.21 mm²**
Dial Gauge Constant:	**0.1 mm per division**
Proving Ring Constant:	**0.05 kN per division**
Loading Rate:	**1.25 mm/min (strain-controlled)**
Standard Reference:	**IS:2720 Part 10**

Mould Preparation

Specimen in UCS Machine

UCS TEST — TRIAL 1 OBSERVATION TABLE

IS:2720 Part 10 · Natural Soil (0% Red Mud) · Specimen: Ø37 mm × 75 mm

A_o = 1075.21 mm² · Dial Gauge Const = 0.1 mm/div · Proving Ring Const = 0.05 kN/div · L_o = 75 mm

Obs.	Dial (div)	PR (div)	ΔL (mm)	Strain ε (%)	Corr. Area Ac (mm²)	Load P (kN)	Stress q (kPa)
1	5	14	0.50	0.667	1082.43	0.70	646.70
2	10	16	1.00	1.333	1089.74	0.80	734.12
3	15	18	1.50	2.000	1097.15	0.90	820.30
4	20	19	2.00	2.667	1104.67	0.95	859.99
5	25	22	2.50	3.333	1112.29	1.10	988.95
6	30	23	3.00	4.000	1120.01	1.15	1026.78
7	35	24	3.50	4.667	1127.84	1.20	1063.98
8★	40	25	4.00	5.333	1135.79	1.25	**1100.56 ★PEAK**
9	45	23	4.50	6.000	1143.84	1.15	1005.38

TRIAL 1 RESULT: qu (UCS) = 1100.56 kPa | Cu (Undrained Shear Strength) = qu / 2 = 550.28 kPa | Peak at Obs. 8 (ε = 5.333%)

UCS TEST — TRIAL 2 OBSERVATION TABLE

IS:2720 Part 10 · Natural Soil (0% Red Mud) · Specimen: Ø37 mm × 75 mm

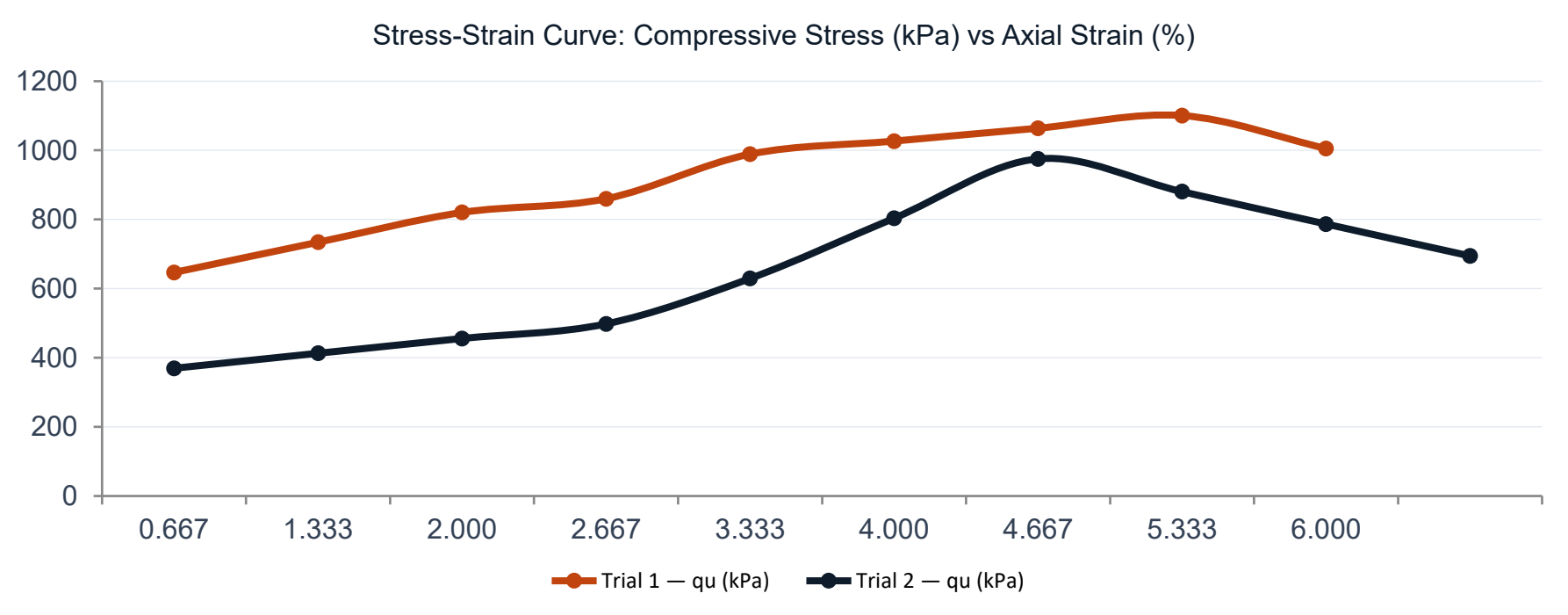
A_o = 1075.21 mm² · Dial Gauge Const = 0.1 mm/div · Proving Ring Const = 0.05 kN/div · L_o = 75 mm

Obs.	Dial (div)	PR (div)	ΔL (mm)	Strain ε (%)	Corr. Area Ac (mm²)	Load P (kN)	Stress q (kPa)
1	5	8	0.50	0.667	1082.43	0.40	369.54
2	10	9	1.00	1.333	1089.74	0.45	412.94
3	15	10	1.50	2.000	1097.15	0.50	455.72
4	20	11	2.00	2.667	1104.67	0.55	497.89
5	25	14	2.50	3.333	1112.29	0.70	629.33
6	30	18	3.00	4.000	1120.01	0.90	803.56
7★	35	22	3.50	4.667	1127.84	1.10	**975.31 ★PEAK**
8	40	20	4.00	5.333	1135.79	1.00	880.45
9	45	18	4.50	6.000	1143.84	0.90	786.82
10	50	16	5.00	6.667	1152.01	0.80	694.44

TRIAL 2 RESULT: qu (UCS) = 975.31 kPa | Cu = 487.66 kPa | Peak at Obs. 7 (ε = 4.667%)

UCS — STRESS-STRAIN CURVES (BOTH TRIALS)

IS:2720 Part 10 · Natural Soil · qu vs Axial Strain (%)



Trial 1 — Peak qu	Trial 2 — Peak qu	Average qu	Average Cu
1100.56 kPa	**975.31 kPa**	**1037.93 kPa**	**518.97 kPa**
at ε = 5.333% (Obs. 8)	*at ε = 4.667% (Obs. 7)*	*(T1 + T2) / 2*	*= Average qu / 2*

UCS TEST — SUMMARY OF RESULTS

Natural Soil (0% Red Mud) · IS:2720 Part 10

Trial 1 qu

1100.56 kPa

Peak at Obs. 8 · ε = 5.333%

Trial 1 Cu

550.28 kPa

= qu / 2

Trial 2 qu

975.31 kPa

Peak at Obs. 7 · ε = 4.667%

Trial 2 Cu

487.66 kPa

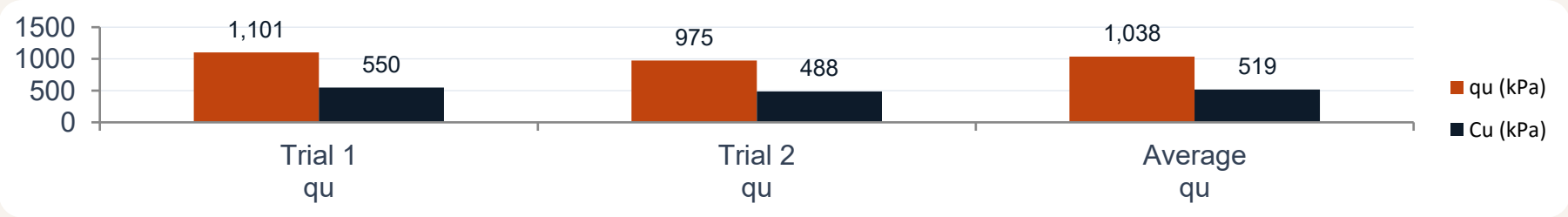
= qu / 2

AVERAGE UCS (qu)

1037.93 kPa

AVERAGE Cu (Shear Strength)

518.97 kPa



ADVANTAGES & APPLICATIONS

Industrial Alkali Waste (Red Mud) as Soil Stabilizer

01

Strength Improvement

Red Mud introduces Fe/Al/Ca oxides causing pozzolanic and cementitious reactions → higher CBR and unconfined compressive strength

02

Reduced Plasticity

Alkaline environment (pH 10–13) flocculates clay particles → lower Liquid Limit (LL), Plasticity Index (PI) and swelling potential

03

Cost-Effective

Industrial waste material — zero raw material cost. Reduces expensive lime or cement stabilization costs by 40–60%

04

Eco-Friendly & Sustainable

Beneficial reuse of hazardous alkali waste → reduces Red Mud dumping, groundwater contamination and land degradation

FIELD APPLICATIONS

- Road Subgrade (Weak highway & rural roads)
- Embankments (Dam fill & road embankments)
- Foundation Soil (Ground improvement for footings)
- Agricultural Land (Soil amendment & pH adjustment)

PENDING WORK & EXPECTED OUTCOMES

Test Matrix Status — Red Mud Proportions: 0%, 5%, 10%, 15%, 20%

Mix Proportion	UCS Status	UCS Result (qu & Cu)	Proctor Status	Proctor Result (OMC & MDD)
0% (Natural Soil)	**✓ Complete**	**qu = 1037.93 kPa · Cu = 518.97 kPa**	**✓ Complete**	**OMC = 21.66% · MDD = 1.539 g/cc**
5% Red Mud	⏳ Pending	—	⏳ Pending	—
10% Red Mud	⏳ Pending	—	⏳ Pending	—
15% Red Mud	⏳ Pending	—	⏳ Pending	—
20% Red Mud	⏳ Pending	—	⏳ Pending	—

EXPECTED OUTCOMES (Based on Literature)

UCS Improvement

Expected 15–35% strength gain at optimum Red Mud %. Pozzolanic reactions cementite soil matrix.

Optimal ~10–15%

Literature shows maximum strength gain typically at 10–15% Red Mud. Beyond this, strength reduces.

Plasticity Reduction

LL and PI expected to reduce due to high alkalinity causing clay particle flocculation.

Waste Utilization

Each tonne of Red Mud reused saves ₹2,000–₹5,000 in disposal cost and prevents land pollution.

FUTURE SCOPE & RESEARCH DIRECTIONS

Extending This Study — Long-Term & Field Applications

01

Complete Test Matrix

Conduct UCS and Standard Proctor tests for all Red Mud proportions: 5%, 10%, 15%, 20%. Plot qu vs % and γd vs w% for each mix.

02

Atterberg Limits Testing

Determine Liquid Limit (LL), Plastic Limit (PL) and Plasticity Index (PI) for each mix. Confirm plasticity reduction with Red Mud.

03

CBR (California Bearing Ratio)

Conduct CBR tests to evaluate subgrade strength for road pavement design. Compare CBR improvement vs natural soil baseline.

04

Combined Stabilization

Combine Red Mud with lime, cement, fly ash or GGBS for synergistic effects. Investigate dual-waste stabilization systems.

05

Free Swell Index Test

Quantify swelling potential reduction with Red Mud addition. Important for expansive clay soils in construction.

06

Field Pilot & Durability Study

Large-scale field testing on actual road subgrade. Study long-term durability under traffic loading, wetting-drying cycles and weathering.

CONCLUSIONS

C1 Fine-grained soil from Ganga River plains classified as weak subgrade requiring stabilization treatment.

C2 Standard Proctor Test (IS:2720 Part 7) gave OMC = 21.66% and MDD = 1.539 g/cc for natural soil.

C3 UCS testing (IS:2720 Part 10) on natural soil: Average qu = 1037.93 kPa, Cu = 518.97 kPa (baseline).

C4 Red Mud (industrial alkali waste) from Bayer process contains Fe_2O_3, Al_2O_3, SiO_2, CaO — proven pozzolanic & cementitious properties.

C5 Addition of Red Mud expected to significantly improve qu, reduce LL/PI and improve γd — optimal % at 10–15% per literature.

C6 This study presents a dual-benefit approach: improving weak soil performance while sustainably utilizing hazardous industrial waste.

THANK YOU

Questions & Discussion Welcome

OMC / MDD	Avg qu (UCS)	Avg Cu
21.66% / 1.539 g/cc	**1037.93 kPa**	**518.97 kPa**

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Project Website / References

https://sauravkutri.github.io/minor_project/